

Chapter 7: Sampling Distributions

Sections 7.1–7.4

MAT 1272 Statistics

Today's Goals

By the end of this lesson, you should be able to:

- 1 Explain what a sampling distribution is.
- 2 Distinguish sampling error from nonsampling error.
- 3 Compute the mean of the sampling distribution of $\bar{x}$.
- 4 Compute the standard error of $\bar{x}$.
- 5 Use the Central Limit Theorem.
- 6 Use the TI-84 to find probabilities involving sample means.

Population Distribution vs Sampling Distribution

Population Distribution

Describes the values of individual observations in the population.

Sampling Distribution

Describes the values of a sample statistic, such as the sample mean $\bar{x}$, from all possible samples of the same size.

x = individual value $\bar{x}$ = sample mean

Example: Small Population

Suppose the population consists of five exam scores:

70, 78, 80, 80, 95

The population mean is

$$\mu = \frac{70 + 78 + 80 + 80 + 95}{5} = 80.60$$

The population standard deviation is approximately

$$\sigma = 8.09$$

Population Probability Distribution

x	f	$P(x)$
70	1	0.20
78	1	0.20
80	2	0.40
95	1	0.20

This distribution describes individual scores.

It is a population distribution because it uses all values in the population.

Now Take Samples

Suppose we take samples of size

$$n = 3$$

from the population.

Each sample has its own sample mean:

$$\bar{x} = \frac{\text{sum of sample values}}{3}$$

The sample mean changes from sample to sample.

Therefore, $\bar{x}$ is a random variable.

Sampling Distribution of $\bar{x}$

Definition

The sampling distribution of $\bar{x}$ is the probability distribution of all possible values of the sample mean $\bar{x}$.

Example values of $\bar{x}$ from samples of size 3:

76.00, 76.67, 79.33, 81.00, 81.67, 84.33, 85.00

Sampling Distribution Table

$\bar{x}$	$P(\bar{x})$
76.00	0.20
76.67	0.10
79.33	0.10
81.00	0.10
81.67	0.20
84.33	0.20
85.00	0.10

This table describes possible sample means, not individual values.

Key Difference

Population Distribution

x

Individual observations.

Examples:

70, 78, 80, 95

Sampling Distribution

$\bar{x}$

Sample means.

Examples:

76.00, 81.67, 85.00

The sampling distribution usually has less spread.

Sampling Error

Sampling Error

Sampling error is the difference between the sample statistic and the corresponding population parameter.

For the sample mean:

$$\text{Sampling error} = \bar{x} - \mu$$

Sampling error happens because different random samples give different sample means.

Example of Sampling Error

Population mean:

$$\mu = 80.60$$

Sample:

$$70, \quad 80, \quad 95$$

Sample mean:

$$\bar{x} = \frac{70 + 80 + 95}{3} = 81.67$$

Sampling error:

$$\bar{x} - \mu = 81.67 - 80.60 = 1.07$$

Nonsampling Error

Nonsampling Error

Nonsampling errors are errors that occur during collection, recording, or processing of data.

Examples:

- ▶ Entering a wrong number
- ▶ Asking a biased survey question
- ▶ Respondents giving false answers
- ▶ Recording or typing mistakes

Mean of the Sampling Distribution

Important Formula

The mean of the sampling distribution of $\bar{x}$ is equal to the population mean.

$$\mu_{\bar{x}} = \mu$$

This means that $\bar{x}$ is an unbiased estimator of μ .

Standard Deviation of $\bar{x}$

The standard deviation of the sampling distribution of $\bar{x}$ is called the standard error.

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

This formula is used when

$$\frac{n}{N} \leq 0.05$$

or when the population is very large.

What Does Standard Error Mean?

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

As n increases, $\sqrt{n}$ increases.

Therefore:

$\sigma_{\bar{x}}$ decreases

Larger samples produce sample means that are closer to the population mean.

Example: Wages of Employees

Suppose

$$\mu = 27.50$$

and

$$\sigma = 3.70$$

For samples of size

$$n = 75$$

we have

$$\mu_{\bar{x}} = 27.50$$

and

$$\sigma_{\bar{x}} = \frac{3.70}{\sqrt{75}} = 0.427$$

Effect of Sample Size

Using

$$\sigma_{\bar{x}} = \frac{3.70}{\sqrt{n}},$$

we get:

n	$\sigma_{\bar{x}}$
30	0.676
75	0.427
200	0.262

As the sample size increases, the spread of the sampling distribution decreases.

Shape of the Sampling Distribution

The shape of the sampling distribution depends on:

- ① The shape of the population distribution.
- ② The sample size n .

There are two main cases:

- ▶ The population is normally distributed.
- ▶ The population is not normally distributed.

Case 1: Population Is Normal

If the population is normally distributed, then the sampling distribution of $\bar{x}$ is also normal. This is true for any sample size n .

$$\bar{x} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

So we can use normal probabilities for $\bar{x}$.

Case 2: Population Is Not Normal

If the population is not normal, the sample size matters.

For small samples:

$$\bar{x}$$

may not be normally distributed.

For large samples:

$$\bar{x}$$

becomes approximately normal.

Central Limit Theorem

Central Limit Theorem

If the sample size is large, usually

$$n \geq 30,$$

then the sampling distribution of $\bar{x}$ is approximately normal, regardless of the shape of the population distribution.

This is one of the most important results in statistics.

Why the Central Limit Theorem Matters

Many real populations are not normal.

Examples:

- ▶ Income
- ▶ Rent
- ▶ Waiting times
- ▶ House prices
- ▶ Medical costs

But with large samples, sample means behave approximately normally.

Example: Rents Paid by Tenants

Suppose the mean rent in a city is

$$\mu = 1550$$

and the standard deviation is

$$\sigma = 225$$

The population distribution is skewed right.

For

$$n = 30,$$

the sampling distribution of $\bar{x}$ is approximately normal by the Central Limit Theorem.

Rent Example: Standard Error

For

$$\mu = 1550, \quad \sigma = 225, \quad n = 30,$$

we have

$$\mu_{\bar{x}} = 1550$$

and

$$\sigma_{\bar{x}} = \frac{225}{\sqrt{30}} = 41.08$$

So

$$\bar{x} \sim N(1550, 41.08)$$

approximately.

Larger Sample Size

If

$$n = 100,$$

then

$$\sigma_{\bar{x}} = \frac{225}{\sqrt{100}} = 22.50$$

Compare:

n	$\sigma_{\bar{x}}$
30	41.08
100	22.50

The larger sample has less variability.

Applications of Sampling Distributions

Once we know

$$\mu_{\bar{x}}$$

and

$$\sigma_{\bar{x}},$$

we can find probabilities involving sample means.

Examples:

$$P(\bar{x} < a)$$

$$P(\bar{x} > a)$$

$$P(a < \bar{x} < b)$$

Empirical Rule for Sample Means

For a normal sampling distribution:

68.26%

of sample means fall within

$$\mu \pm 1\sigma_{\bar{x}}$$

95.44%

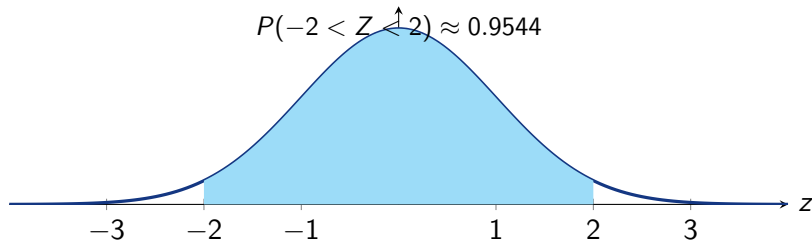
fall within

$$\mu \pm 2\sigma_{\bar{x}}$$

99.74%

fall within

Visual: Within Two Standard Errors



$$P(\mu - 2\sigma_{\bar{x}} \leq \bar{x} \leq \mu + 2\sigma_{\bar{x}}) \approx 0.9544$$

TI-84 Important Note

Students already know how to standardize:

$$z = \frac{\bar{x} - \mu}{\sigma_{\bar{x}}}$$

But the TI-84 can compute the area directly using:

$$\mu_{\bar{x}} \quad \text{and} \quad \sigma_{\bar{x}}$$

Use:

`normalcdf(lower, upper, mean, standard deviation)`

Example: Packages of Cookies

Suppose cookie package weights are approximately normal with

$$\mu = 32$$

ounces and

$$\sigma = 0.3$$

ounces.

A random sample of

$$n = 20$$

packages is selected.

Find:

$$P(31.8 < \bar{x} < 31.9)$$

Cookie Example: Standard Error

$$\mu_{\bar{x}} = 32$$

$$\sigma_{\bar{x}} = \frac{0.3}{\sqrt{20}} = 0.0671$$

So we use:

$$\bar{x} \sim N(32, 0.0671)$$

TI-84 command:

```
normalcdf(31.8,31.9,32,0.0671)
```

Cookie Example: Result

The calculator gives approximately

$$0.0667$$

Therefore,

$$P(31.8 < \bar{x} < 31.9) \approx 0.0667$$

Interpretation:

There is about a 6.67% chance that the sample mean weight is between 31.8 and 31.9 ounces.

Example: Checking Accounts

Suppose the annual cost of checking accounts has

$$\mu = 400$$

and

$$\sigma = 30$$

A sample of

$$n = 225$$

accounts is selected.

Find:

$$P(396 \leq \bar{x} \leq 404)$$

Checking Account Example

Compute the standard error:

$$\sigma_{\bar{x}} = \frac{30}{\sqrt{225}} = 2$$

So

$$\bar{x} \sim N(400, 2)$$

TI-84 command:

```
normalcdf(396,404,400,2)
```

Result:

0.9544

Why Does the Answer Look Familiar?

The interval is

$$396 \leq \bar{x} \leq 404$$

This is the same as

$$400 - 2(2) \leq \bar{x} \leq 400 + 2(2)$$

So the interval is within two standard errors of the mean.

$$P(\mu - 2\sigma_{\bar{x}} \leq \bar{x} \leq \mu + 2\sigma_{\bar{x}}) = 0.9544$$

Another Checking Account Probability

Find:

$$P(\bar{x} \leq 397.30)$$

Using the same information:

$$\mu_{\bar{x}} = 400$$

$$\sigma_{\bar{x}} = 2$$

TI-84 command:

```
normalcdf(-1E99,397.30,400,2)
```

Result:

0.0885

General Strategy

For probability questions involving $\bar{x}$:

- 1 Identify μ , σ , and n .
- 2 Compute

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}.$$

- 3 Decide whether the sampling distribution is normal or approximately normal.
- 4 Use `normalcdf` with mean μ and standard deviation $\sigma_{\bar{x}}$.

When Can We Use NormalCDF for $\bar{x}$?

We can use normal methods when:

- ▶ The original population is normal; or
- ▶ The sample size is large:

$$n \geq 30.$$

Then the sampling distribution of $\bar{x}$ is normal or approximately normal.

Practice Problem 1

A population has

$$\mu = 80, \quad \sigma = 12$$

A sample of size

$$n = 36$$

is selected.

Find:

- ① $\mu_{\bar{x}}$
- ② $\sigma_{\bar{x}}$
- ③ $P(\bar{x} < 83)$

Practice Problem 1: Calculator Setup

$$\mu_{\bar{x}} = 80$$

$$\sigma_{\bar{x}} = \frac{12}{\sqrt{36}} = 2$$

For

$$P(\bar{x} < 83),$$

use:

`normalcdf(-1E99,83,80,2)`

Practice Problem 2

A population has

$$\mu = 50, \quad \sigma = 8$$

A sample of size

$$n = 16$$

is selected from a normal population.

Find:

$$P(48 < \bar{x} < 53)$$

Since the population is normal, the sampling distribution of $\bar{x}$ is normal even though $n < 30$.

Practice Problem 2: Calculator Setup

$$\sigma_{\bar{x}} = \frac{8}{\sqrt{16}} = 2$$

Use:

`normalcdf(48,53,50,2)`

This directly computes

$$P(48 < \bar{x} < 53)$$

without manually converting to z-scores.

Chapter 7 Summary

Key Formulas

$$\mu_{\bar{x}} = \mu$$

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

$$z = \frac{\bar{x} - \mu}{\sigma_{\bar{x}}}$$

TI-84 shortcut:

`normalcdf(lower, upper, mean, standard deviation)`

What You Should Know

You should be able to:

- 1 Explain the meaning of a sampling distribution.
- 2 Distinguish sampling error and nonsampling error.
- 3 Compute standard error.
- 4 Apply the Central Limit Theorem.
- 5 Use `normalcdf` for sample mean probabilities.